

Anthony M. Nguyen

US Citizen | (619) 455-4983 | minhansd@gmail.com | [LinkedIn](#) | [Website \(minhansd.github.io\)](http://Website(minhansd.github.io))

SUMMARY AND INTERESTS

Electrical and Computer Engineering Ph.D. student at University of Michigan with machine learning and robotics research experience. Interested in control theory, linear systems, multi-agent systems, safety, reinforcement learning, distributed optimization, state estimation, robotics, and machine learning.

SKILLS

Relevant Coursework: Robotic Sensing and Estimation, Robotic Planning and Learning, Linear Systems, Control Theory, Optimization, Probability Theory, Random Processes, Machine Learning, Signal Processing
Programming Languages: Python (*NumPy*, *PyTorch*, *JAX*, *TensorFlow*), MATLAB, C, Java, Verilog, VHDL
Tools: MATLAB, QuestaSim, Active-HDL, Xilinx Vivado, LaTeX/Overleaf, RedHat Linux

EDUCATION

University of Michigan (UMich) Ann Arbor, MI
Ph.D., Electrical and Computer Engineering (Advisor: Necmiye Ozay) August 2026 - May 2031 (expected)

University of California, San Diego (UCSD) La Jolla, CA
B.S., Electrical Engineering (GPA: 3.982/4.000; Magna Cum Laude) September 2023 - June 2026

EXPERIENCE

Existential Robotics Laboratory (UCSD) La Jolla, CA
Undergraduate Student Researcher January 2025 – Present

- Developed Monte Carlo tree search (MCTS) algorithms for optimal control of partially observable Markov decision processes (POMDPs); Successfully implemented in Python for sensor scheduling problems
- Discovered, formalized, and proved relationships between different approaches to particle flow particle filtering (PFPF); Developed and proved methods to avoid numerical instability for PFPF
- Implemented PFPF methods in Python for linear and nonlinear observation models using analytical and numerical methods; Used ordinary differential equation (ODE) solvers (*SciPy*, *DiffraX*) and GPU-acceleration libraries (*JAX*)

L3Harris Technologies, Inc. Carlsbad, CA
Electrical Engineering Intern June 2025 – September 2025

- Studied and presented on latest generative model and robotics technologies to the team
- Migrated a communication design in Verilog from Altera to Xilinx technology; Updated designs for the company's new hardware; Recreated transmission flagging logic on the new design

Cognicom, Inc. San Diego, CA
Electrical Engineering Intern January 2023 – September 2023

- Implemented in Verilog serial communication protocols, created a precision timer circuit, and created a Xilinx Zynq bare-metal C program to control an LCD screen for a tactical radio

UC San Diego Institute of Engineering in Medicine (IEM): OPALS La Jolla, CA
Research Intern June 2022 – August 2022

- Worked to implement and analyze UCSD Biophotonics Lab research papers relating to cell microsurgery robots
- Worked with a graduate student to test results for a research project about DNA repair in cells treated with red light

AWARDS

Magna Cum Laude (UCSD) June 2026

- Top 4% in GPA of all graduating students in the class of 2026 at UCSD.

ECE Booker Memorial Honors Award (UCSD) May 2026

- Awarded to students in UCSD ECE who graduate with a GPA above 3.7.

Provost Honors (UCSD) Fall 2023 – Spring 2026

- Awarded to at the end of each quarter to students who achieve a 3.5 or higher UC GPA in at least 12 graded units

George Eastman Young Leaders Award (High School) May 2022

- Awarded for strong leadership, high grades and challenging courses, and extensive extracurricular activities